

# AeroMind

## AI-Powered Airport Disruption Command Centre

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### Project Planning Documentation

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### Background:

Airport operations generate thousands of flight movements daily. Delays, cancellations, congestion, and operational disruptions can propagate across airports and airlines, creating significant financial and operational impacts. Most airport operators rely on historical reporting and reactive decision-making rather than predictive intelligence.

### Problem Statement:

Airport operators struggle to identify future disruptions, understand delay propagation, and proactively manage congestion before operational performance deteriorates.

### Project Objective:

Develop an AI-powered decision intelligence platform that predicts operational disruptions, explains root causes, forecasts future risks, and recommends corrective actions.

### Business Goals:

- Improve disruption visibility
- Predict future operational risks
- Identify root causes of delays
- Support proactive decision-making
- Demonstrate AI-assisted airport analytics

### Expected Deliverables:

- Enterprise Data Model
- SQL Analytics Layer
- Python Analytics Pipeline
- Machine Learning Models
- AI Recommendation Engine
- Power BI Executive Dashboards
- Project Documentation

### Primary Stakeholders:

- Airport Operations Manager
- Airport CEO
- Airline Operations Team
- Ground Operations Team
- Airport Analytics Team

**Technology Stack:**

- Python
- SQL
- Power BI Desktop
- XGBoost
- Isolation Forest
- Ollama
- Llama 3
- CrewAI

**Expected Outcome:**

A production-style airport operations intelligence platform capable of predicting disruptions and generating AI-assisted operational recommendations.